THE COMPLETION OF AN AFFINE SCHEME ALONG A CLOSED SUBSET IS A MCQUILLAN AFFINE FORMAL ALGEBRAIC SPACE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one colimit description of the functor; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- A subset of the spectrum of a ring is closed exactly when it is the vanishing locus of an ideal, and vanishing on all primes means nilpotence. Algebra, Lemma algebra-lemma-Zariski-topology, invoked at proof lines 3163-3165. Turns the condition that every prime contains the extension of the radical ideal into nilpotence of its elements.
- The defining condition for a McQuillan affine formal algebraic space. Same chapter, Lemma lemma-mcquillan-affine-formal-algebraic-space at proof lines 3174-3176, and Definition definition-types-affine-formal-algebraic-space at proof lines 3173-3174. Gives the McQuillan property from surjectivity of the maps from the completion.
- Affine formal algebraic spaces as filtered colimits of affine schemes. Same chapter, Definition definition-affine-formal-algebraic-space, in force in the statement. Fixes the shape of the presentation produced.

Source reference. Original lemma 00E0: sources/stacks-chow/tags.txt, source lines 517–517. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIC: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2135–2135. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 07E8: sources/stacks-chow/tags.txt, source lines 9438–9438. Pointer outside the counted span, invoked once for context; the

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Source reference. Original definition 0GXE: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1595–1595. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. Let X be an affine formal algebraic space over S. We say X is *McQuillan* if X satisfies the equivalent conditions of Lemma 0AIC. Let A be the weakly admissible topological ring associated to X. We say

(1) X is *classical* if X is McQuillan and A is admissible (More on Algebra, Definition 07E8),
(2) X is *weakly adic* if X is McQuillan and A is weakly adic (Definition 0GXE),
(3) X is *adic* if X is McQuillan and A is adic (More on Algebra, Definition 07E8),
(4) X is *adic** if X is McQuillan, A is adic, and A has a finitely generated ideal of definition, and
(5) X is *Noetherian* if X is McQuillan and A is both Noetherian and adic.

Lemma 0.2. *Let S be a scheme. Let X be an affine scheme over S. Let $T \subset |X|$ be a closed subset. Then the functor*

$$(Sch/S)_{fppf} \longrightarrow Sets, \quad U \longmapsto \{f : U \to X \mid f(|U|) \subset T\}$$

is a McQuillan affine formal algebraic space.

Proof. Say $X = \mathrm{Spec}(A)$ and T corresponds to the radical ideal $I \subset A$. Let $U = \mathrm{Spec}(B)$ be an affine scheme over S and let $f : U \to X$ be an element of $F(U)$. Then f corresponds to a ring map $\varphi : A \to B$ such that every prime of B contains $\varphi(I)B$. Thus every element of $\varphi(I)$ is nilpotent in B, see Algebra, Lemma 00E0. Setting $J = \mathrm{Ker}(\varphi)$ we conclude that I/J is a locally nilpotent ideal in A/J. Equivalently, $V(J) = V(I) = T$. In other words, the functor of the lemma equals $\mathrm{colim}\,\mathrm{Spec}(A/J)$ where the colimit is over the collection of ideals J with $V(J) = T$. Thus our functor is an affine formal algebraic space. It is McQuillan (Definition 0.1) because the maps $A \to A/J$ are surjective and hence $A^\wedge = \lim A/J \to A/J$ is surjective, see Lemma 0AIC. □